

NodeX LAN Chat

Real-time local network messaging using ESP8266 microcontrollers



Project Overview

What is NodeX LAN Chat?

A real-time local network messenger built using ESP8266 microcontrollers. It creates its own WiFi hotspot and allows multiple devices to chat without internet connectivity.

This project combines embedded systems programming with modern web development to create a seamless messaging experience.

Key Highlights

- WebSocket-powered real-time communication
- Modern dark UI with smooth animations
- Emoji support with built-in picker
- Message reactions and typing indicators



Problem Statement



No Internet Required

Enables communication in areas without internet access or cellular coverage



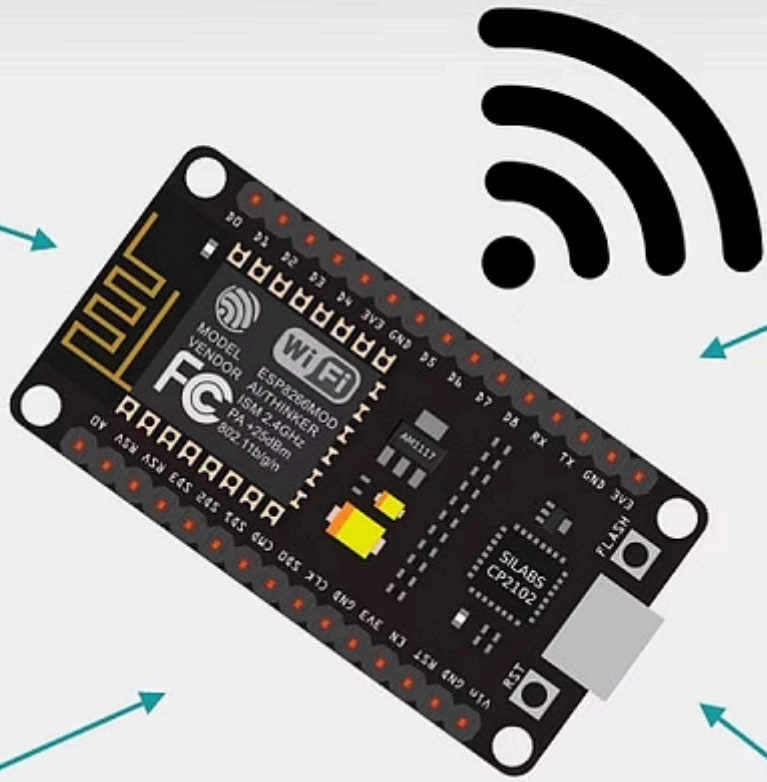
Local & Secure

Messages stay within your local network, ensuring privacy and security



Group Communication

Facilitates instant messaging among multiple users on the same network



ESP8266 ACCESS POINT

The Solution



ESP8266 Server

Microcontroller creates WiFi hotspot and hosts chat server



Web Interface

Users access chat through browser at 192.168.4.1



Real-time Updates

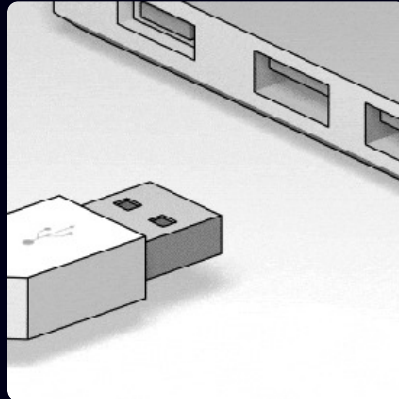
WebSocket technology enables instant messaging without refresh

Hardware Requirements



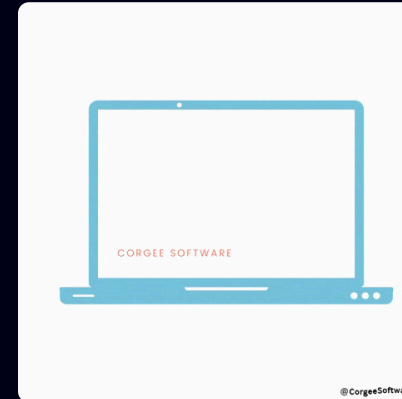
ESP8266 Module

NodeMCU or ESP-12E development board with built-in WiFi capabilities



USB Cable

Micro USB cable for programming and power supply



Programming Device

Laptop or computer with Arduino IDE for uploading code

Software Requirement

1

Arduino IDE

Main development environment for writing and uploading code to ESP8266

2

ESP8266WiFi Library

Enables WiFi connectivity and hotspot creation functionality

3

ESP8266WebServer

Hosts the web interface and serves HTML, CSS, and JavaScript files

4

WebSocketsServer

Provides real-time bidirectional communication between server and clients

5

ArduinoJson

Handles JSON data parsing for message formatting and transmission

Key Features



Real-time Chat

Instant messaging using WebSockets with no page refresh required



Emoji Support

Built-in emoji picker with search functionality for expressive communication



Delete Messages

Remove your messages for everyone in the chat



Group & Private Chat

Communicate with everyone or send one-to-one private messages



Message Reactions

React to messages using 👍 😂 😮 😢 🔥 100 🐱 emojis



Typing Indicator

See when someone is currently typing a message

Advanced Features

Auto Message History

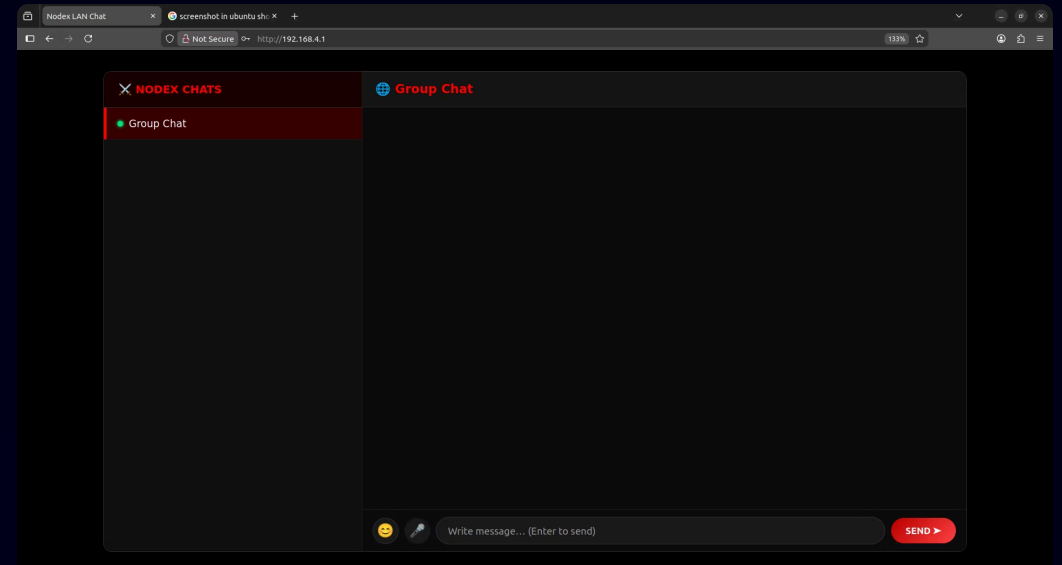
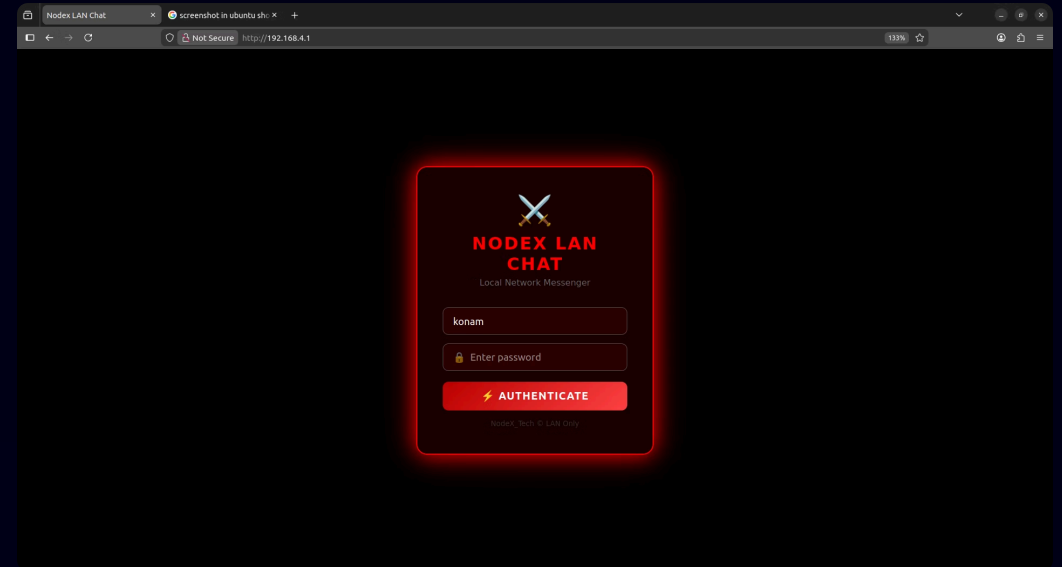
Restores last 45 seconds of messages when new users join the chat

Ghost Mode

Secret feature that allows sending anonymous messages

Sound Effects

Audio feedback for sending, receiving, joining, and deleting messages





How to Use



Upload Code

Open Arduino IDE, select board as NodeMCU 1.0 (ESP-12E Module), and upload NodeX_Chats_final.ino to your ESP8266



Connect to WiFi

Power on the ESP8266. Connect your device to WiFi network named "NodeX_Tech" with password "password"



Open Chat Interface

Launch browser and navigate to <http://192.168.4.1/>



Login

Enter login password: 112211, choose your username, and start chatting!